AST 6007-01 Celestial Mechanics

Midterm Examination

October 20, 2020 10:10-12:10

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In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) The nonlinear differential equations that describe the dynamics of a spacecraft under the influence of the gravity of the Earth and propulsive forces are expressed in spherical coordinates as follows:

$$\ddot{r} - r\dot{\nu}^2 \cos^2\lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + \frac{F_r}{m}$$

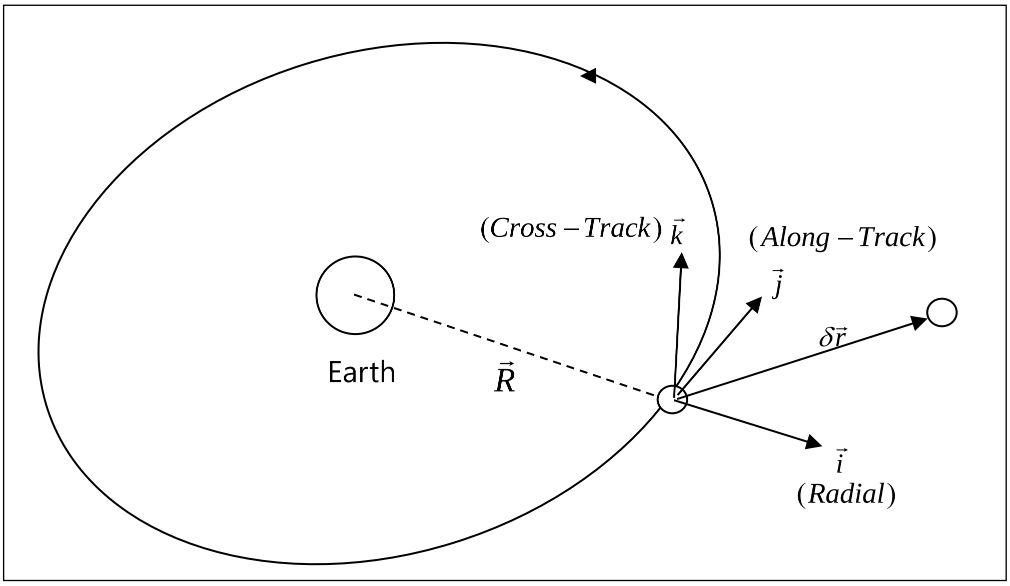
$$r\ddot{\nu}\cos\lambda + 2\dot{r}\dot{\nu}\cos\lambda - 2r\dot{\nu}\dot{\lambda}\sin\lambda = \frac{F_\nu}{m}$$

$$r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2\sin\lambda\cos\lambda = \frac{F_\lambda}{m}$$

Here (r, ν, λ) represents a triple of (radial distance from the Earth center, longitude, latitude), (F_r, F_ν, F_λ) represents a triple of force components in the associated direction, m is the mass of the spacecraft, and $\mu = 4\times10^5 km^3/\sec^2$ is the gravitational parameter of the Earth.

A. (5) Suppose that the spacecraft is in a circular orbit of $(r \equiv 7{,}000km, \lambda \equiv 0)$ with no propulsive forces acting on it. What are the constant orbital velocity and orbital period of the spacecraft?

B. (15) Linearize with respect to an arbitrary circular reference orbit to derive the Hill-Clohessy-Wiltshire equations.



2. (40) Consider a satellite initially on a geosynchronous circular equatorial orbit whose constant orbital frequency is denoted as $\vec{\omega} = \omega\vec{k}$. After performing a maneuver, it is now located at the following relative position and velocity with respect to its non-maneuvering position in the local-vertical-local-horizontal (LVLH) $\vec{i}\vec{j}\vec{k}$ -frame attached to the geosynchronous orbit:

$$(x, y, z) = (-110, 70, 10)km$$
$$(\dot{x}, \dot{y}, \dot{z}) = (-10, 20, 1)m/\sec$$

If the satellite is to initiate a two-impulse maneuver to return to its original non-maneuvering location in 2 hours, what are the changes in velocities at the initial and terminal time, i.e., $\Delta\vec{v}_0$ and $\Delta\vec{v}_f$?

Hint: The solution to the Hill-Clohessy-Wiltshire (HCW) equation is given as

$$x(t) = \frac{\dot{x}_0}{\omega}\sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right)\cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right)$$
$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right)\sin(\omega t) + \frac{2\dot{x}_0}{\omega}\cos(\omega t) - \left(6\omega x_0 + 3\dot{y}_0\right)t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$
$$z(t) = \frac{\dot{z}_0}{\omega}\sin(\omega t) + z_0\cos(\omega t)$$

If the relative position in the HCW equation is equal to zero after time *t*, the initial position and velocity must satisfy the following relation:

$$\dot{y}_0 = \frac{\left[6x_0(\omega t - \sin(\omega t)) - y_0\right]\omega\sin(\omega t) - 2\omega x_0\left(4 - 3\cos(\omega t)\right)(1 - \cos(\omega t))}{\left(4\sin(\omega t) - 3\omega t\right)\sin(\omega t) + 4\left(1 - \cos(\omega t)\right)^2}$$
$$\dot{x}_0 = -\frac{\omega x_0\left(4 - 3\cos(\omega t)\right) + 2\left(1 - \cos(\omega t)\right)\dot{y}_0}{\sin(\omega t)}$$
$$\dot{z}_0 = -z_0\omega COT(\omega t)$$

3. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

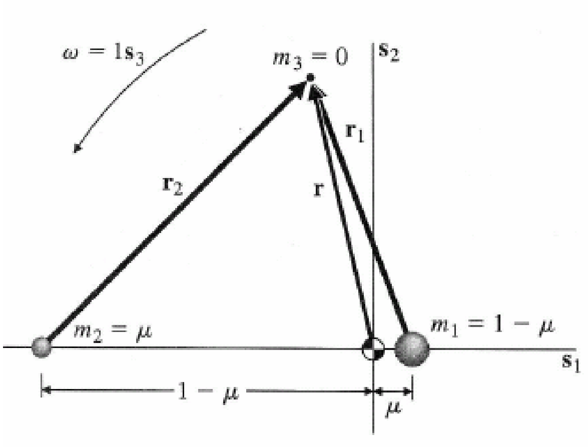


FIGURE 10.1
Geometry of the restricted problem.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$
$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$
$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$
$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2},\ r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$
$\mu = 0.0123$ for the Earth-Moon System

A. (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?

B. (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame?

4. (20) Consider a two-dimensional linear system

$$\Delta\dot{x} = A\Delta x,\ \ \Delta x(0) = \Delta x_0$$

Let $\lambda = \pm j\omega$ and $(\xi, \bar{\xi})$ be the eigenvalues and eigenvectors of the system, respectively. Describe how to find the initial conditions that compose a Halo Orbit with a single frequency of ω.